

■ **SENSOR:** Accelerometer · Gyroscope · Camera (MoveNet pose)

The problem

People rarely notice their gait, balance or left-right symmetry deteriorating until it causes pain, injury or a fall. Clinical gait/balance labs cost tens of thousands and need specialists — unavailable in most homes and rural areas.

How it works

- Short walk / balance / Timed-Up-&-Go test drives the IMU + camera.
- Accelerometer & gyroscope give cadence, stride time, trunk sway and step-time asymmetry.
- Real MoveNet pose tracking adds posture & symmetry from the camera.
- A personal baseline is stored; future tests are compared to it (drift detection).
- On-device AI coach explains the numbers and builds a corrective-exercise plan.

Example output

Left-right gait asymmetry: 8%

Trunk stability drops after 10 min (fatigue signature)

Balance worsened 12% vs your baseline — consider a check-up

Cadence 164 spm · stride consistency: good

Metrics

Stability score · L-R asymmetry % · trunk sway · cadence · consistency · fatigue drift

Novelty & positioning

High — sensor fusion (IMU + camera pose) is rare in consumer apps; the shared 2026 Frontiers paper is only conceptual.

Target users

Runners · fitness users · elderly (fall risk) · rehab patients · general population

Assistive / 'cure' feature

Corrective exercise plan (single-leg balance, tandem stance, core, cadence drills) — most gait/balance issues improve with practice.

Research backing

- Smartphone accelerometer gait: ICC 0.78–0.99 vs lab (MDPI Appl. Sci. 2024)
- Single-camera BlazePose running gait: ICC 0.75–0.98 vs Vicon (Sensors 2023)
- Shared paper — smartphone vestibular & gait framework (Front. Digit. Health 2026)

On-device AI coach

Personal baseline + drift

History & charts

Monthly PDF report

Contact-a-specialist

100% private / offline

Part of the **SenseKit** suite (MotionSense · SpeakSense · HearSense · VisionSense). Screening & self-monitoring only — not a medical diagnosis. Phase 2 = native app with calibrated sensors.